

FORECAST-TO-RANK ALLOCATION (F2R)

System Architecture and Operating Framework

A machine-learning cross-asset strategy that converts forecasts into relative ranks and a monthly Top-4 portfolio

SYSTEM DOCUMENTATION v1.0

This document explains how Forecast-to-Rank Allocation is structured, how forecasts are converted into portfolio decisions, and how those decisions are executed, measured, and released. It is system documentation, not a replication manual, research paper, or trading instruction.

SlackQuant Research

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Document Summary

Field	Details
Document	Forecast-to-Rank Allocation (F2R): System Architecture and Operating Framework
Version	1.0
Release date	2026-09-09
System type	Portfolio Strategy System
Public identity	Forecast-to-Rank Allocation (F2R) · Machine-Learning Cross-Asset Portfolio Strategy
Live information set	Five price-derived features from each ETF's own history; macroeconomic inputs are not used by the current live strategy
Forecast layer	Four-model ensemble: Ridge, Elastic Net, Random Forest, and XGBoost
Portfolio rule	Equal-weight cross-asset rank ensemble; Top-4 assets at 25% each
Decision and execution cycle	Monthly decision after month-end; execution at the first subsequent common U.S. trading-day close
Performance measurement	Continuous daily return series under the live accounting convention; risk statistics annualized using 252 trading days
Live transaction cost	10 bp per side; turnover measured from drifted pre-trade weights
Research boundary	Research evidence remains separately identified and retains the cost convention used in the original study

How this document is organized

Sections 1-8 explain the strategy, information set, forecast-and-rank process, portfolio construction, timing, and accounting. Sections 9-13 cover performance, evidence, PDS integration, operations, and disclosure. Sections 14-15 cover validation and related methods; the appendices summarize status terms, resources, limitations, and revision triggers.

Terminology

In this document, F2R means Forecast-to-Rank Allocation.

OVERVIEW

1. Executive Overview

Forecast-to-Rank Allocation (F2R) is a monthly multi-asset portfolio strategy. Each model produces an asset-level forecast; F2R ranks the 11 assets within each model, combines the resulting ranks across models, and allocates 25% to each of the four highest-ranked assets.

The current live strategy is price-based. It uses five price-derived features calculated from each ETF's own history and does not require macroeconomic data. Macro-only and macro-plus-price variants remain part of the preserved research record, but they are not part of the current live decision rule.



What defines F2R

- An 11-ETF cross-asset universe based on each ETF's actual trading history.
- Asset-level forecasts generated by four current model families.
- Cross-sectional ranking within each model, followed by equal-weight aggregation across models.
- A Top-4 equal-weight target portfolio produced on a monthly decision cycle.
- A defined execution schedule, drift-aware turnover calculation, and bilateral transaction-cost policy.
- Clear separation among the Official Decision, Intramonth Preview, Current MTD, and preserved research evidence.

What F2R is not

- The current live strategy is not a macro-timing strategy; macro variants belong to the research record.
- Forecast magnitude does not map directly to portfolio weight.
- The Intramonth Preview is provisional; it does not authorize execution or enter performance accounting.
- The historical operating series is a reconstruction under the live accounting convention, not a claim of contemporaneous live management throughout the sample.

Core distinction

Machine learning estimates relative opportunity; it does not set portfolio weights directly. F2R translates forecasts into ranks, combines those ranks across models, and applies a fixed Top-4 equal-weight rule.

ARCHITECTURE

2. System Architecture

F2R separates forecasting from portfolio construction. The forecast layer estimates next-period return separately for each asset. The portfolio layer ranks those forecasts cross-sectionally and applies fixed selection and weighting rules. Forecast magnitude therefore does not determine portfolio weight directly.

Stage	Step	What happens
01	Dated market input	Use only ETF history available by the signal cutoff.
02	Feature construction	Calculate the five live price features for each ETF.
03	Forecasting	Fit the four current model families using training pairs whose subsequent returns are already observable.
04	Within-model ranking	Within each model, rank the 11 asset forecasts from highest to lowest.
05	Rank ensemble	Convert ranks to common percentiles and average them equally across the four models.
06	Portfolio formation	Select the Top-4 and assign 25% target weight to each selected ETF.
07	Execution and accounting	Rebalance at the first subsequent common U.S. trading-day close, apply turnover and modeled costs, and continue daily accounting.

Why the separation matters

A larger forecast does not mechanically receive a larger weight. Portfolio weights are determined by cross-sectional order and the fixed Top-4 rule. The decision can therefore be traced through distinct forecast, ranking, selection, target, execution, and performance stages.

Missing data do not create a new state

F2R does not invent missing inputs or fill gaps with synthetic forecasts, targets, execution dates, or performance values. A new state is accepted only after its source, timing, and consistency checks pass.

UNIVERSE AND INPUTS

3. Investment Universe and Live Information Set

The current live F2R universe contains 11 U.S.-listed ETFs covering U.S. equity styles, international equities, real assets, and U.S. Treasury duration. Each asset is modeled from its own ETF price history. F2R does not extend the live universe with proxy series during normal operation.

Ticker	Exposure	Live universe
SPY	U.S. large-cap equity	Included
QQQ	U.S. large growth / Nasdaq-100	Included
IWM	U.S. small-cap equity	Included
EFA	Developed markets ex-U.S. equity	Included
EEM	Emerging-markets equity	Included
VNQ	U.S. listed real estate	Included
DBC	Broad commodities	Included
IEF	Intermediate U.S. Treasuries	Included
TLT	Long U.S. Treasuries	Included
EWY	Korea equity	Included
GLD	Gold	Included

Five price features used in live forecasting

Input feature	Definition
1-month return	Trailing 1-month total return
3-month return	Trailing 3-month cumulative total return
6-month return	Trailing 6-month cumulative total return
12-month return	Trailing 12-month cumulative total return
12-month volatility	Realized volatility calculated from trailing monthly returns

Macro data are not part of the live input set

The current live F2R strategy uses only each ETF's price history. Macroeconomic series, including FRED-MD variables, are not used to produce the Official Decision or Intramonth Preview. Research experiments that use macro data remain separately identified and do not alter the live specification.

FORECAST LAYER

4. Forecasting Models and Estimation

For each ETF and forecast date, F2R trains four model families on an expanding historical sample. A training observation enters the sample only after the return it is meant to predict is fully known. At least 120 such observations are required. Whenever model selection is needed, validation preserves time order rather than using randomly shuffled folds.

Model	Forecast role	Estimation and validation
Ridge	Linear L2-regularized model	Predictors are standardized within the training sample. Time-ordered validation selects the penalty; the model is then refit on the full sample available at that forecast date.
Elastic Net	Linear L1/L2-regularized model	Predictors are standardized within the training sample. The mixing ratio is fixed; time-ordered validation selects the penalty; the model is then refit.
Random Forest	Bagged nonlinear tree model	Fixed live configuration with a deterministic seed; no month-by-month hyperparameter search.
XGBoost	Boosted nonlinear tree model	Chronological validation with early stopping; the model is refit on the full sample using the selected number of boosting rounds.

Information controls in model training

- Input features use only prices available by the signal cutoff.
- A labeled training observation is used only after the subsequent return it predicts is fully known; for signal month t , the live Preview construction therefore ends the labeled sample at $t-2$.
- Model-selection steps preserve time order; randomly shuffled cross-validation is not used in the live specification.
- Scaling parameters are estimated within the training sample available at that forecast date, not from the full historical dataset.
- Stochastic models use deterministic seeds tied to the model, asset, forecast horizon, and forecast date.

Why use four models?

F2R does not choose a single model to dominate the portfolio each month. The four models provide different forecasting structures; F2R combines their relative rankings so the portfolio decision reflects agreement across model types rather than one model's forecast scale.

PORTFOLIO TRANSLATION

5. Forecast-to-Rank Ensemble and Top-4 Allocation

Forecasts from different model families are not directly comparable in scale. F2R therefore converts each model's forecasts into a cross-asset ranking before combining the four models.



Ranking procedure

- For each signal month and model, the 11 assets are ordered from highest to lowest predicted return.
- Each asset is assigned a rank and a rank percentile on a common 0-to-1 scale.
- The four model rank percentiles are averaged with equal weight across models.
- If assets remain tied, the mean model z-score is used first; a predefined asset order provides the final tie-break.
- The final ordering determines the portfolio: ranks 1-4 are selected and receive 25% each.

Target weight = 25% for final ranks 1–4; 0% for ranks 5–11

What ranking changes

Ranking emphasizes relative opportunity rather than the absolute size of a forecast. Forecasts with very different magnitudes can receive neighboring ranks when they occupy neighboring positions within their model. This makes the four model outputs comparable, but target weights depend on ordering rather than forecast magnitude.

Fixed portfolio rule

The Top-4 equal-weight rule is part of the live specification. Research on alternative models, universes, or ensembles does not change the Official portfolio unless the live specification is formally updated.

TIMING

6. Decision and Execution Timing

F2R uses a monthly information and execution clock. The signal is formed after the final common U.S. market close of the completed signal month. The target is executed at the first subsequent common U.S. trading-day close. The portfolio already in place earns through that rebalance close; the new target begins earning from the next close-to-close interval.



Monthly timing rules

- The signal month uses only market history available through its final common U.S. market close.
- The signal is not assumed executable at the same close that supplied the final information.
- Execution occurs at the first subsequent common U.S. trading-day close across the 11-ETF universe.
- Pre-trade weights reflect market-driven drift accumulated since the previous rebalance.
- Realized performance is assigned to the calendar month in which it occurs, not backdated to the signal month.

Clock item	Interpretation
Signal month	Completed month whose information produces the target
Execution date	First subsequent common U.S. trading-day close
Holding return	Return earned by the portfolio in force after the relevant rebalance
Calendar performance	Aggregation of the continuous daily return series

Why signal formation and execution use different closes

Using the month-end close both to form the signal and to assume execution would create an execution-timing inconsistency. F2R therefore separates the signal cutoff from the rebalance close.

ACCOUNTING

7. Portfolio Accounting, Weight Drift, and Transaction Costs

F2R rebalances to target weights monthly rather than resetting weights every day. Between rebalances, asset weights move with market returns. At the next rebalance, turnover is measured from those drifted pre-trade weights.

Turnover

$$\text{One-way turnover} = 0.5 \times \sum | \text{target weight} - \text{drifted pre-trade weight} |$$

The factor of one half avoids counting the same portfolio replacement once as a sale and again as a purchase.

Live cost convention

$$\text{Live transaction cost} = 2 \times \text{one-way turnover} \times 0.001$$

The live convention is 10 basis points per side. A complete 100% one-way portfolio replacement therefore implies 20 basis points of modeled transaction cost.

Research cost convention

Preserved research evidence retains the convention used in the original study: 10 basis points applied to one-way turnover. Those historical results are not restated under the later live bilateral convention, so the two cost definitions remain explicitly separated.

Context	Turnover basis	Cost treatment
Live operating series	Drifted pre-trade half-L1	10 bp per side; bilateral
Preserved research evidence	Drifted pre-trade half-L1	10 bp one-way; original research convention

Keep research and live cost conventions separate

Restating historical research with a later cost convention would change the original evidence. F2R therefore preserves the research results as reported and applies the current bilateral convention only to the live operating series.

PORTFOLIO STATUS

8. Official Decision, Intramonth Preview, and Current MTD

F2R presents several views of the current state, but they do not carry the same authority. A more recent timestamp does not by itself make a view actionable.

View	What it is	Execution / performance authority
Official Decision	Monthly Top-4 target designated as the current Official Decision	Yes; execution follows the monthly decision and execution clock
Intramonth Preview	Provisional target based on completed common closes during the current signal month	No; target-only and may change before month-end
Current MTD	Month-to-date mark of the Official portfolio through the latest completed common U.S. market close	Monitoring only; calculated from Official holdings, never Preview holdings
Research Evidence	Preserved historical study results and robustness analyses	Evidence only; does not replace the live Official Decision

Preview discipline

- Preview uses the same live price-only information set, model families, ranking rule, and Top-4 mapping as the Official Decision process.
- Preview may use the latest completed common close within the month, but it does not create an execution, turnover event, or realized performance record.
- If a Preview cannot be produced, it is shown as unavailable rather than interpreted as an empty portfolio or a full exit.
- Current MTD is calculated from Official holdings whether or not a Preview is available.

Authority rule

Only the Official Decision has strategy decision authority. Intramonth Preview and Current MTD are monitoring views; they cannot replace or change the Official Decision on their own.

PERFORMANCE

9. Performance Measurement

The primary F2R operating performance view is a continuous daily return series reconstructed under the current live accounting convention. It uses validated historical target records together with subsequent Official targets and is kept separate from the original research return series.

Measure	Primary operating definition
Support	Daily operating series begins 2017-05-01 and extends as validated market data and target history become available
Monthly return	Calendar-month aggregation of the daily net-return series
Annual return	Calendar-year aggregation of monthly returns
Volatility	Sample standard deviation of daily returns $\times \sqrt{252}$
Sharpe	Mean daily return / daily standard deviation $\times \sqrt{252}$; risk-free rate assumed to be zero
Drawdown	Peak-to-trough decline in the daily wealth series
CAGR	Terminal wealth annualized over elapsed calendar time

Why a daily base matters

Monthly, annual, cumulative, volatility, Sharpe, drawdown, and CAGR measures all come from the same daily return series. This keeps operating performance internally consistent rather than combining monthly research returns with a separate daily series.

Consistency with PDS

As part of the current validation, standalone F2R monthly operating returns were compared with the F2R provider series used by PDS at 24 anchor months. The maximum absolute difference was approximately 0.0023 basis points, within the adjusted-price precision tolerance.

How to interpret the historical operating series

The reconstructed history applies the current live accounting convention to validated historical targets. It does not imply that F2R was operated live throughout the full reconstructed period.

EVIDENCE BOUNDARY

10. Research Evidence and the Current Live Specification

F2R developed within a broader forecasting and allocation research program. The current live system uses a narrower specification than the full set of research experiments, and the two are kept clearly separate.

Layer	Current role	Examples
Current live F2R specification	Produces the Official Decision and Intramonth Preview targets	Own-price features; four-model rank ensemble; native 11-ETF universe; Top-4 equal weight
Preserved primary research	Original evidence and study record	112 primary signal origins; original one-way research cost convention; matched benchmark studies
Macro research branches	Research evidence only	Macro-only and macro-plus-own information sets; timing and vintage robustness
Alternative universes and robustness tests	Research comparison only	Same-universe and expanded-universe comparability tests; not automatically adopted
Future system research	Exploration until separately validated and adopted	Alternative ML models, time-series foundation models, ensembles, or portfolio mappings

Why price-only is live

The current live strategy uses only the own-price information set. Macro variants remain part of the research program because they address a different research question; they are not inputs to the live F2R decision.

What remains preserved

- Original research performance and validation records are not retrospectively rewritten to match later live conventions.
- Live maintenance may correct accounting or data-maturity implementation issues without changing the original research record.
- Experimental models remain outside the live system until they are separately validated and formally adopted.

Research record remains separate

This documentation release does not alter the original research evidence or validation record. Changes to live implementation, documentation, or presentation are identified separately.

SYSTEM RELATIONSHIP

11. Relationship to Portfolio Decision System (PDS)

F2R is an independent Portfolio Strategy System. PDS operates at the portfolio level, where it combines admitted strategy providers into an integrated portfolio. The two systems use aligned execution-clock conventions so their accounting remains comparable, while each retains its own decision authority, research record, and operating responsibility.

Dimension	F2R	PDS
Primary role	Generate the strategy-level multi-asset target	Combine admitted strategy targets into an integrated portfolio under PDS policy
Decision ownership	Owns F2R forecasts, ranks, Top-4 Official Decision, Intramonth Preview, and strategy accounting	Owns provider integration, sleeve weights, the PDS portfolio decision, monitoring, and public disclosure
Current relationship	One of the current strategy providers admitted to the PDS Active Core	Integrates F2R with ADAA under the current Core policy
Execution clock	First subsequent common close; existing holdings earn through the rebalance close; the new target earns from the next interval	Uses aligned execution timing when provider decisions are integrated into portfolio accounting
Research ownership	F2R retains ownership of its research evidence	Provider evidence remains provider evidence; PDS does not relabel it as PDS evidence

What PDS does not do to F2R

- PDS does not refit or replace the F2R forecasting models.
- PDS does not create an F2R target when the provider output is unavailable.
- PDS does not turn an F2R Intramonth Preview into an Official Decision.
- PDS may add portfolio-level integration costs without double-counting costs already reflected by F2R.

Hierarchy

F2R answers “what should this strategy hold?” PDS answers “how should admitted strategy providers be combined and monitored as one portfolio?”

OPERATIONS

12. Operating and Publication Process

F2R uses one controlled operating workflow. Refresh, validation, publication approval, and deployment are separate steps, so rebuilding local state does not publish anything by itself.



Normal operator actions

Action	Purpose
Refresh + rebuild + open	Refresh the Official Decision, Preview, and accounting state; rebuild local dashboard inputs; then open the dashboard.
Refresh + publish	Refresh and validate current state, build the public release package, show the publication delta, require explicit approval, then update the deployment repository.
System Status	Run validation gates and report the local, validated, current-public, and repository state.
Advanced / recovery	Access refresh-only, publication preparation, audit, setup, logs, and controlled recovery functions.

Three-state release model

- Latest Local State: newest locally rebuilt state that has passed the required source checks.
- Validated State: public release package that has passed publication validation.
- Current Public State: validated release currently published through the deployment repository.

Publication is not automatic

Preparation builds and validates a public release package. Publication verifies the same package, shows what will change, requires explicit operator approval, and only then updates the deployment repository.

PUBLIC DISCLOSURE

13. Public Disclosure and Deployment Boundary

The F2R public dashboard can show the current Official Decision, Intramonth Preview, Current MTD, and approved historical evidence. These are strategy-level F2R outputs. Their publication does not extend to account holdings, private operating state, or protected PDS portfolio information.

Publicly visible	Not publicly disclosed
Formal F2R identity and strategy architecture	Internal scientific and runtime identifiers
Current Official Decision	Local filesystem paths, credentials, and environment state
Intramonth Preview, explicitly provisional	Research branches and experimental outputs not adopted into the live system
Current MTD based on Official holdings	Private validation and release files
Approved historical operating series and target history	PDS or provider information outside F2R's public scope
Related research and public deployment repository	Any state that has not passed the required timing, source, and release checks

Public website and deployment repository

The SlackQuant System page provides the F2R overview and links to this documentation. The standalone Streamlit application provides the public dashboard, and its deployment repository contains the public application files. These surfaces report validated F2R outputs; they do not determine the strategy state. Operational decisions remain with the F2R system.

A public target is not an execution record

An Official F2R target is a strategy decision. It is not evidence that a broker order was filled, nor is it a custody or account-level record of actual holdings.

RELEASE CHECKS

14. Release Validation

Before v1.0 was released, the live information set, model ensemble, portfolio rule, execution timing, cost treatment, performance series, status boundaries, and public disclosure were checked against the current operating records. The table summarizes the main checks. It describes the system at release; it is not a forecast of future performance.

Check	Status	What was confirmed
Public identity	PASS	Forecast-to-Rank Allocation (F2R) name and public system role matched the current platform
Live information set	PASS	Own-price feature set confirmed; macro data are not inputs to the live Official Decision or Intramonth Preview
Model ensemble	PASS	Ridge, Elastic Net, Random Forest, and XGBoost confirmed in the current live ensemble
Rank / portfolio rule	PASS	Equal model-rank combination; final Top-4 assets receive 25% each
Execution clock	PASS	First subsequent common U.S. trading-day close; prior portfolio earns through the rebalance close
Turnover / live cost	PASS	Half-L1 turnover from drifted pre-trade weights; 10 bp per side
Performance path	PASS	Continuous daily operating series; daily risk statistics annualized using 252 trading days
Preview boundary	PASS	Preview is target-only and does not create execution, turnover, or realized performance
Research boundary	PASS	Original research evidence and its one-way cost convention remain separately preserved
PDS consistency	PASS	24-anchor monthly comparison remained within the adjusted-price precision tolerance
Public boundary	PASS	Public release excludes local/private paths and internal state outside the approved public scope

What this validation means

These checks confirm that v1.0 describes the current live specification and public boundary. They do not promote a new forecasting model, change an Official Decision, alter the preserved research record, or create new investment evidence.

RELATED METHODS

15. Related Quantitative Methods

The SlackQuant Quantitative Methods library provides general background for methods used in F2R or needed to interpret its validation and performance. The articles explain the underlying concepts; this System Documentation explains how those concepts are applied in the current F2R system.

Method	Topic	Relevance to F2R
QM001	Out-of-Sample Forecast Evaluation	Defines the out-of-sample information boundary that applies before forecasts are converted into ranks.
QM002	Rolling vs Expanding Windows	Provides context for the expanding training sample used by current F2R and for alternative estimation-window choices.
QM003	Look-Ahead Bias and Data Leakage	Explains the timing and model-development controls required for valid historical forecast evaluation.
QM007	Portfolio Backtesting and Rebalancing	Explains monthly targets, execution timing, weight drift, and holding-period accounting.
QM009	Turnover, Transaction Costs, and Net Performance	Explains drift-aware turnover and how modeled transaction costs affect net performance.
QM013	Benchmark-Relative Portfolio Evaluation	Explains active return, tracking error, information ratio, and other benchmark-relative measures.

Methods library: research.slackquant.com/methods/

How to read these references

Use the Methods library for general concepts and definitions. Use this System Documentation for F2R-specific choices, including the expanding training sample, cross-asset ranking, execution timing, transaction costs, and the authority of each public status view.

F2R-specific implementation details

If a general Methods article and the current F2R implementation differ on an implementation detail, the current F2R specification controls. The Methods article remains conceptual background.

LIMITATIONS

Appendix A. Limitations and Revision Policy

Limitations

- Forecast models can lose predictive value when market relationships change; using several model families does not eliminate regime risk.
- Cross-asset ranking intentionally uses relative ordering rather than the full magnitude of each forecast.
- A Top-4 portfolio can be concentrated relative to the 11-ETF universe and can experience meaningful turnover when rankings change.
- The 11-ETF universe is a deliberate design choice and does not represent the full investable opportunity set.
- Transaction costs are modeled assumptions and may differ from actual spreads, market impact, taxes, financing costs, or broker charges.
- The historical operating series is reconstructed under the current execution clock and cost convention; it does not imply that F2R was operated live throughout the full historical period.
- Current MTD is a monitoring estimate until the calendar month is complete and the result is validated.
- Intramonth Preview can change before month-end and has no execution or realized-performance authority.
- Research on macro information, alternative universes, or alternative models does not change the current live specification unless separately validated and formally adopted.

When a documentation version changes

Change	Version treatment
Presentation-only edit	Normally unchanged unless the edit changes meaning.
Routine monthly state or data update	No version change; the operating record is refreshed.
Live information set, model ensemble, or ranking rule	New documentation version required.
Universe, Top-N, or target-weight rule	New documentation version required.
Execution timing, turnover, cost, or performance-accounting treatment	New documentation version required.
Public disclosure policy	New documentation version required.

Versioning rule

A new documentation version is issued when a material change affects the live information set, model ensemble, portfolio rule, execution timing, cost or performance accounting, or public disclosure policy. Routine monthly updates and presentation-only edits do not require a new version unless their meaning changes.

STATUS AND RESOURCES

Appendix B. Decision, Evidence, and Publication Status

Item	Authority	Can change the Official Decision?	Used in realized performance?	Public?
Official Decision	Current strategy decision	Yes	Yes, after execution under the live accounting convention	Yes
Intramonth Preview	Provisional target-only view	No	No	Yes
Current MTD	Month-to-date monitoring view based on Official holdings	No	Monitoring only; not a completed performance record	Yes
Historical operating series	Reconstructed daily operating history	No	Yes, for completed historical performance reporting	Yes
Preserved research evidence	Original study evidence	No	Research evidence only	Yes where approved
Macro and robustness research	Research only	No	No live performance authority	Selective, for context
Public deployment copy	Validated copy for publication	No	No independent authority	Yes

Primary public resources

Resource	Location
F2R System page	research.slackquant.com/systems/f2r/
F2R Public Dashboard	f2r-forecast-to-rank-allocation.streamlit.app
System Documentation	research.slackquant.com/resources/systems/f2r/F2R_System_Documentation_v1.0.pdf
Related Research	research.slackquant.com/research/price-macro-decision/
Public deployment repository	github.com/slackquant80/f2r-forecast-to-rank-allocation
Quantitative Methods	research.slackquant.com/methods/

Suggested reading order

Start with the F2R System page for the architecture overview, use the Public Dashboard for the current public strategy state, and use this document for the operating rules and evidence boundaries. The related research provides the broader empirical study.

RELEASE SUMMARY

Appendix C. F2R v1.0 Summary

Item	F2R v1.0
Public name	Forecast-to-Rank Allocation (F2R)
System type	Portfolio Strategy System
Live universe	11 U.S.-listed ETFs
Live features	1M, 3M, 6M, 12M returns and 12M realized volatility
Forecast models	Ridge, Elastic Net, Random Forest, XGBoost
Forecast combination	Equal-weight cross-sectional model-rank ensemble
Portfolio rule	Top-4 equal weight; 25% per selected asset
Decision cycle	Monthly
Execution	First subsequent common U.S. trading-day close after signal month-end
Weight evolution	Market-driven drift between monthly rebalances
Live turnover	Half-L1 versus drifted pre-trade weights
Live cost	10 bp per side; bilateral
Primary operating performance	Continuous daily operating series; daily risk statistics annualized using 252 trading days
Research performance	Preserved separately; original one-way research cost convention retained
Intramonth Preview	Intramonth, target-only, non-execution
Current MTD	Provisional month-to-date mark based on Official holdings; Preview excluded
PDS relationship	Independent strategy provider currently integrated into the PDS Active Core
Research record	Original research evidence and validation record preserved separately

About this version

This version describes the F2R framework in effect on 9 September 2026. Earlier research and development variants remain part of the project history, but they are not used to describe the current live system.

SlackQuant Research

System documentation only. Not investment advice.